

COURSE CODE	COURSE TITLE	L	T	P	TW &SL	C
UMA3352	FOUNDATIONS OF DISCRETE MATHEMATICS (Computer Science Engineering) (Third semester)	45	15	0	60	4

OBJECTIVES

- To equip the students with the concepts of classical logic, normal forms and its applications.
- To solve problems using counting techniques and graph theory
- To study the algebraic structures and Boolean algebra and their applications.

Syllabus

UNIT I MATHEMATICAL LOGIC 12

Introduction to logic: Multivalued logic – three-valued logic, connectives – Two valued logic: Propositions – Propositional equivalences – Normal forms - Predicates and quantifiers – Nested quantifiers – Rules of inference – Introduction to proofs – Proof methods and strategy.

UNIT II COMBINATORICS 12

Mathematical induction – The pigeonhole principle – Recurrence relations – Solving linear recurrence relations using generating functions – Principle of Inclusion and Exclusion – Divisibility – Primes – The Chinese remainder theorem.

UNIT III GRAPHS 12

Graphs: basic definitions – special types of graphs – Subgraphs – Matrix representation of graphs – Connectivity – Eulerian and Hamilton graphs – Trees: Properties – Distance and Centres – Spanning Tree.

UNIT IV ALGEBRAIC STRUCTURES 12

Algebraic systems – Semi groups and monoids – Groups–Subgroups – Homomorphisms – Normal subgroup and cosets – Lagrange’s theorem – Rings and Fields: Definitions and examples.

UNIT V LATTICES AND BOOLEAN ALGEBRA 12

Relations – Equivalence relation – Partial ordering – Posets – Lattices as Posets – Properties of lattices – Lattices as algebraic systems – Sublattices – Boolean algebra – Stone’s representation Theorem.

Lecture	: 45 Periods
Tutorial	: 15 Periods
Term work & Self-Learning	: 60 Periods
Total	: 120 Periods

OUTCOMES

On successful completion of this course, the student will be able to

CO1: apply logical inferences in deriving decisions in engineering problems

CO2: formulate and solve problems using recurrence relation and various counting techniques.

CO3: apply graph theoretical techniques in network analysis.

CO4: identify various algebraic structures and solve problems using their properties.

CO5: identify and solve problems using Boolean algebra.

TEXTBOOKS

1. Kenneth H Rosen, “Discrete Mathematics and its Applications”, 8th Edition, Special Indian edition, Tata McGraw Hill, New Delhi, 2021.
2. Tremblay J P and Manohar R, “Discrete Mathematical Structures with Applications to Computer Science”, 35th Reprint, Tata McGraw Hill, New Delhi, 2008.

REFERENCE BOOKS

1. Ralph P Grimaldi, “Discrete and Combinatorial Mathematics: An Applied Introduction”, 4th Edition, Pearson Education Asia, 2007.
2. Eric Lehman, F Tom Leighton, Albert R Meyer, Mathematics for Computer Science, Samurai Media Limited, 2017.
3. Thomas Koshy, “Discrete Mathematics with Applications”, Elsevier Publications, 2004.
4. C.L. Liu, DP Mohapatra, “Elements of Discrete Mathematics: A Computer Oriented Approach”, 4th Edition, McGraw Higher Education, 2017.
5. John M Harris, Jeffery L Hirst, Michael J Mossinghoff, “Combinatorics and Graph Theory”, Springer Verlag New York, 2008.

CO-PO MAPPING

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